

Computing a software project budget

Computing a software project budget involves defining the scope, creating a Work Breakdown Structure (WBS), and estimating labor (hours × rate), infrastructure, and licensing costs.

Use bottom-up estimation for accuracy by summing task-level costs, then add contingency (15–20%+) for risks. Development typically consumes 30–50% of the total budget.

Key Components of a Software Budget

- **Labor Costs (Highest Component):** Salaries for developers, PMs, designers, and testers. Calculate based on estimated hours x hourly rate.
- **Infrastructure:** Cloud services (AWS/Azure), servers, hosting, and domain fees.
- **Software Licensing:** Third-party APIs, tools, and subscriptions.
- **Contingency Buffer:** 15–20% of the total budget to cover unexpected issues.

Steps to Compute the Budget

1. **Define Scope ([WBS](#)):** Break down the project into smallest deliverables.
2. **Estimate Resources:** Identify personnel and tools needed for each deliverable.
3. **Calculate Costs:** Multiply estimated hours by hourly rates for each task.
4. **Add Contingency:** Include a budget for risk mitigation.
5. **Review and Approve:** Validate with stakeholders.

Typical Budget Allocation

- **Development:** 30%–50%
- **Testing/QA:** 15%–20% (up to 35% for complex projects)
- **Documentation:** 10%–25%
- **Project Management:** 5%–20%
- **Design/Prototyping:** 5%–10%

Budgeting Methods

- **[Bottom-Up Estimating](#):** Summing costs for each task (most accurate).
- **[Analogous Estimating](#):** Using data from previous similar projects.
- **[Parametric Estimating](#):** Using statistical modeling (e.g., cost per line of code or hours per feature).

Example: 3-Month Mobile App MVP

This example assumes a mid-sized mobile app development project with a 3-month timeline.

Category	Description	Calculation	Estimated Cost
Labor (Internal)	1 Sr. Developer, 1 QA Engineer, 1 Designer	(480 hrs/person 3 people) \$50/hr	\$72,000
Project Management	1 PM (part-time)	120 total hours \$60/hr	\$7,200
Infrastructure	AWS/Google Cloud hosting, DB services	\$300/mo 3 months	\$900
Software Licenses	IDEs, Design tools, Testing platforms	One-time/Monthly fees	\$1,500
External Services	API integrations (e.g., Maps, Payments)	Usage-based estimates	\$2,000
Subtotal			\$83,600
Contingency (15%)	Buffer for risks/scope changes	\$83,600 0.15	\$12,540
TOTAL BUDGET			\$96,140

Breakdown by Development Phase

Most software projects distribute their total budget across these standard phases:

- **Design & Prototyping (10%):** UI/UX wireframes and mockups.
- **Coding & Development (50%):** Building front-end and back-end logic.
- **Testing & QA (25%):** Bug fixes, performance, and user acceptance.
- **Project Management (10%):** Coordination and stakeholder alignment.
- **Deployment & Initial Support (5%):** App Store submission and server setup.

Example 2: Computing a SaaS (Software as a Service) platform budget

Computing a **SaaS (Software as a Service) platform budget** requires planning for complex recurring expenses beyond the initial build.

A production-ready SaaS MVP typically ranges from **\$50,000 to \$120,000**.

1. SaaS Budget Allocation (Phase-by-Phase)

The total build cost is typically distributed across these standard development stages:

Stage	% of Budget	Typical Cost	Key Deliverables
Discovery & Planning	10–15%	\$5,000 – \$15,000	Requirements (SRS), architecture, market fit
UI/UX Design	10–20%	\$8,000 – \$30,000	Prototypes, design systems, branding
Core Development	40–50%	\$40,000 – \$120,000	Multi-tenancy, APIs, security, backend
Testing & QA	10–15%	\$5,000 – \$25,000	Manual/automated tests, security audits
DevOps & Launch	5–10%	\$8,000 – \$25,000	Cloud setup, CI/CD pipelines, SSL

2. Mandatory "SaaS-Only" Feature Costs

Founders often miss the engineering cost of non-core "essential" SaaS features:

- **Subscription & Billing:** Integrating Stripe or Chargebee with dunning and tax logic costs **\$8,000 – \$30,000**.
- **Security & Compliance:** GDPR, HIPAA, or SOC 2 compliance can add **\$20,000 – \$50,000** to the initial budget.
- **AI Integration:** Adding GenAI capabilities adds **15–30%** to development and **\$500 – \$5,000/mo** in usage fees.

3. Year 1 Operational Budget (Post-Launch)

Launch is only the "appetizer." Annual maintenance typically costs **15–25%** of the original build price.

- **Infrastructure (Cloud):** AWS/Azure/GCP costs scale with users.
 - *Early stage (<1k users):* **\$200 – \$800/mo**.
 - *Growth stage (1k–50k users):* **\$800 – \$5,000/mo**.
- **Marketing & Sales:** Critical for Product-Led Growth (PLG). Plan for **\$5,000 – \$15,000/mo** for early-stage customer acquisition.
- **Ongoing Maintenance:** A team for bug fixes and security patches typically runs **\$10,000 – \$20,000/mo** as the user base grows.

4. Summary of SaaS Scale Costs

Complexity	Cost Range	Timeline
Micro SaaS	\$10,000 – \$25,000	1–3 months
Standard B2B MVP	\$50,000 – \$120,000	4–7 months
Mid-Market Platform	\$120,000 – \$300,000	8–14 months
Enterprise SaaS	\$300,000 – \$600,000+	14–24 months